

Cybersecurity Dashboard: Learning, Planning & Design Document

1. What is a Cybersecurity Dashboard?

A cybersecurity dashboard is a visual tool that displays an organization's security status in real time. It provides insights into threats, system logs, alerts, and user activities to help security teams monitor and respond to incidents promptly.

2. Importance of SIEM (Security Information & Event Management) Tools

SIEM tools are critical in cybersecurity because they:

- Aggregate and analyze security data from multiple sources
- Detect suspicious activities and generate alerts
- Provide forensic data for incident analysis
- Visualize metrics through dashboards and reports
- Automate threat detection and response workflows

3. Example Features of a Cybersecurity Dashboard

Feature	Description
Log Monitoring	Collect and view logs from systems and applications in real time
Threat Detection	Alerts for suspicious behaviors like failed logins, port scans, etc.
Real-Time Analytics	Visual representation of security metrics and event trends
System Health Monitoring	Monitor CPU, RAM, and disk usage of critical systems

4. Planned Features for the Dashboard

A. Log Monitoring

- Collect system and application logs
- Real-time log updates
- Filtering by date, severity, source

B. Threat Detection Alerts

- Real-time alerts for:
- Multiple failed login attempts
- Suspicious file changes

- Unauthorized port access
- Alerts categorized by severity (Low, Medium, High)

C. Analytics and Visualization

- Daily login attempts (bar chart)
- Types of threats detected (pie chart)
- Timeline of incidents (line graph)

D. System Health Check

- CPU, memory, and disk usage monitoring for critical machines

5. **Technology Stack**

Component	Chosen Stack	Justification
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Frontend	React.js	Fast and responsive UI with component-based architecture
Backend	Flask (Python)	Lightweight, easy to set up APIs for log handling and alerts
Database	MongoDB	Flexible, JSON-like structure ideal for log and alert data
Visualization	Recharts / Chart.js	Integrates well with React, good for data-driven dashboards
Log Input	Filebeat	Efficient log shipper to forward logs to the backend or storage